EFSP Auxiliary ESL Curriculum

Engineering English Jargon Quick Reference

Field-specific terms, contrast pairs, and high-pressure sentence frames

Audience: mechanical, electrical, civil, systems, industrial, test, quality, manufacturing, and field engineers, plus engineering managers and technical project leads

Focus: An engineering English curriculum for requirements, design reviews, tradeoffs, testing, failure analysis, quality, safety factors, manufacturability, field issues, and technical disagreement.

Designed for advanced ESL learners who already use professional English and need industry-specific terminology, realistic meetings, role-play pressure, careful pushback, and polished workplace outputs.

Teaching stance: this is language and workplace-communication training, not legal, medical, financial, safety, or regulatory advice. Instructors should connect every scenario to the learner's current company policies, local rules, and approved procedures.

Nomenclature and Jargon

These are classroom working definitions. Learners should adapt wording to their organization's policies, systems, and local regulatory environment.

Requirements and Constraints

Term	Working meaning	Collocations	Contrast and workplace line
requirement	In engineering, requirement is an approved rule or decision framework that sets what is permitted, required, or acceptable. Use it when teams need to clarify requirements before proposing technical solutions.	apply the requirement; comply with the requirement	Contrast: requirement is an approved rule or threshold, not an informal preference. Example: Please confirm which requirement applies before we commit the team.
constraint	In engineering, constraint is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to clarify requirements before proposing technical solutions.	clarify constraint; document constraint	Contrast: constraint should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify constraint and the evidence that supports it.
tolerance	In engineering, tolerance is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to clarify requirements before proposing technical solutions.	clarify tolerance; document tolerance	Contrast: tolerance should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify tolerance and the evidence that supports it.
tradeoff	In engineering, tradeoff is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to clarify requirements before proposing technical solutions.	clarify tradeoff; document tradeoff	Contrast: tradeoff should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify tradeoff and the evidence that supports it.

Design Reviews and Technical Pushback

Term	Working meaning	Collocations	Contrast and workplace line
design review	In engineering, design review is a structured examination of evidence, process, or results used to reach a defensible conclusion. Use it when teams need to challenge assumptions using evidence rather than status.	conduct a design review; document the design review	Contrast: design review is a structured examination of evidence; it is not a conclusion reached before the evidence is checked. Example: The team will conduct a design review before it recommends a corrective action.
design rationale	In engineering, design rationale is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to challenge assumptions using evidence rather than status.	track design rationale; explain a change in design rationale	Contrast: design rationale measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track design rationale by segment and explain what changed.
risk	In engineering, risk is a condition or event that can create harm, loss, noncompliance, delay, or unacceptable performance. Use it when teams need to challenge assumptions using evidence rather than status.	assess risk; mitigate risk	Contrast: risk identifies a possible or current exposure; it is not the same as accepting that exposure. Example: We need to assess risk, name the owner, and agree on the trigger for escalation.

Term	Working meaning	Collocations	Contrast and workplace line
verification	In engineering, verification is a repeatable control mechanism used to prevent, detect, or confirm conditions before work proceeds. Use it when teams need to challenge assumptions using evidence rather than status.	clarify verification; document verification	Contrast: verification should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify verification and the evidence that supports it.

Failure Modes and Reliability

Term	Working meaning	Collocations	Contrast and workplace line
failure mode	In engineering, failure mode is a condition or event that can create harm, loss, noncompliance, delay, or unacceptable performance. Use it when teams need to explain failure risk, severity, and detection.	clarify failure mode; document failure mode	Contrast: failure mode should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify failure mode and the evidence that supports it.
FMEA	Failure mode and effects analysis, a structured method for identifying how a process or product can fail and prioritizing controls.	clarify FMEA; document FMEA	Contrast: FMEA should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify FMEA and the evidence that supports it.
reliability	In engineering, reliability is a performance or governance dimension that must be protected through defined evidence, thresholds, and accountability. Use it when teams need to explain failure risk, severity, and detection.	clarify reliability; document reliability	Contrast: reliability should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify reliability and the evidence that supports it.
duty cycle	In engineering, duty cycle is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain failure risk, severity, and detection.	clarify duty cycle; document duty cycle	Contrast: duty cycle should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify duty cycle and the evidence that supports it.

Testing, Validation, and Data Interpretation

Term	Working meaning	Collocations	Contrast and workplace line
prototype	In engineering, prototype is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss test results without overgeneralizing.	clarify prototype; document prototype	Contrast: prototype should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify prototype and the evidence that supports it.
acceptance criteria	In engineering, acceptance criteria is an approved rule or decision framework that sets what is permitted, required, or acceptable. Use it when teams need to discuss test results without overgeneralizing.	apply the acceptance criteria; comply with the acceptance criteria	Contrast: acceptance criteria is an approved rule or threshold, not an informal preference. Example: Please confirm which acceptance criteria applies before we commit the team.
validation	In engineering, validation is a repeatable control mechanism used to prevent, detect, or confirm conditions before work proceeds. Use it when teams need to discuss test results without overgeneralizing.	clarify validation; document validation	Contrast: validation should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify validation and the evidence that supports it.

Term	Working meaning	Collocations	Contrast and workplace line
sample size	In engineering, sample size is a defined information asset or analytical construct used to measure, explain, or predict an outcome. Use it when teams need to discuss test results without overgeneralizing.	clarify sample size; document sample size	Contrast: sample size should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify sample size and the evidence that supports it.

Manufacturability and Cost Engineering

Term	Working meaning	Collocations	Contrast and workplace line
DFM	In engineering, DFM is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to connect design decisions to production and cost reality.	clarify DFM; document DFM	Contrast: DFM should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify DFM and the evidence that supports it.
tooling	In engineering, tooling is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to connect design decisions to production and cost reality.	clarify tooling; document tooling	Contrast: tooling should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify tooling and the evidence that supports it.
yield	In engineering, yield is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to connect design decisions to production and cost reality.	improve yield; analyze yield loss	Contrast: Yield measures conforming output, not the root cause of a process change. Example: The fab is analyzing yield loss by tool, layer, and time window.
cycle time	In engineering, cycle time is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to connect design decisions to production and cost reality.	clarify cycle time; document cycle time	Contrast: cycle time should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify cycle time and the evidence that supports it.

Safety Factors and Compliance

Term	Working meaning	Collocations	Contrast and workplace line
safety factor	In engineering, safety factor is a performance or governance dimension that must be protected through defined evidence, thresholds, and accountability. Use it when teams need to explain safety margin and regulatory implications clearly.	clarify safety factor; document safety factor	Contrast: safety factor should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify safety factor and the evidence that supports it.
code compliance	In engineering, code compliance is a performance or governance dimension that must be protected through defined evidence, thresholds, and accountability. Use it when teams need to explain safety margin and regulatory implications clearly.	clarify code compliance; document code compliance	Contrast: code compliance should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify code compliance and the evidence that supports it.

Term	Working meaning	Collocations	Contrast and workplace line
margin	In engineering, margin is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to explain safety margin and regulatory implications clearly.	track margin; explain a change in margin	Contrast: margin measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track margin by segment and explain what changed.
liability	In engineering, liability is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain safety margin and regulatory implications clearly.	clarify liability; document liability	Contrast: liability should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify liability and the evidence that supports it.

Field Issues and Customer Communication

Term	Working meaning	Collocations	Contrast and workplace line
field failure	In engineering, field failure is a condition or event that can create harm, loss, noncompliance, delay, or unacceptable performance. Use it when teams need to communicate technical problems to customers without speculation.	clarify field failure; document field failure	Contrast: field failure should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify field failure and the evidence that supports it.
warranty	In engineering, warranty is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to communicate technical problems to customers without speculation.	clarify warranty; document warranty	Contrast: warranty should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify warranty and the evidence that supports it.
corrective action	Action taken to fix a current problem and prevent recurrence.	clarify corrective action; document corrective action	Contrast: corrective action should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify corrective action and the evidence that supports it.
installation condition	In engineering, installation condition is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to communicate technical problems to customers without speculation.	clarify installation condition; document installation condition	Contrast: installation condition should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify installation condition and the evidence that supports it.

Systems Integration and Interface Control

Term	Working meaning	Collocations	Contrast and workplace line
interface control	In engineering, interface control is a repeatable control mechanism used to prevent, detect, or confirm conditions before work proceeds. Use it when teams need to manage cross-disciplinary dependencies.	clarify interface control; document interface control	Contrast: interface control should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify interface control and the evidence that supports it.
integration	In engineering, integration is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to manage cross-disciplinary dependencies.	track integration; explain a change in integration	Contrast: integration measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track integration by segment and explain what changed.

Term	Working meaning	Collocations	Contrast and workplace line
regression test	In engineering, regression test is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to manage cross-disciplinary dependencies.	clarify regression test; document regression test	Contrast: regression test should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify regression test and the evidence that supports it.
configuration management	In engineering, configuration management is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to manage cross-disciplinary dependencies.	track configuration management; explain a change in configuration management	Contrast: configuration management measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track configuration management by segment and explain what changed.

Industry-Specific Meeting Moves

Situation	Useful language
Requirements and Constraints	Before we commit, I want to confirm requirement, constraint, the owner, and the evidence behind the decision. If load, tolerance, material, manufacturability, regulatory, and cost constraints conflict., I recommend we document the risk and agree on the next step.
Design Reviews and Technical Pushback	Before we commit, I want to confirm design review, design rationale, the owner, and the evidence behind the decision. If design rationale, risk, analysis, and verification evidence should be reviewed., I recommend we document the risk and agree on the next step.
Failure Modes and Reliability	Before we commit, I want to confirm failure mode, FMEA, the owner, and the evidence behind the decision. If operating envelope, duty cycle, failure mode, and reliability target matter., I recommend we document the risk and agree on the next step.
Testing, Validation, and Data Interpretation	Before we commit, I want to confirm prototype, acceptance criteria, the owner, and the evidence behind the decision. If test conditions, sample size, acceptance criteria, and failure analysis are incomplete., I recommend we document the risk and agree on the next step.
Manufacturability and Cost Engineering	Before we commit, I want to confirm DFM, tooling, the owner, and the evidence behind the decision. If yield, tooling, cycle time, supplier capability, and cost must be balanced., I recommend we document the risk and agree on the next step.
Safety Factors and Compliance	Before we commit, I want to confirm safety factor, code compliance, the owner, and the evidence behind the decision. If safety factor, code requirements, test evidence, and liability exposure need review., I recommend we document the risk and agree on the next step.
Field Issues and Customer Communication	Before we commit, I want to confirm field failure, warranty, the owner, and the evidence behind the decision. If evidence, installation conditions, warranty terms, and corrective action are not complete., I recommend we document the risk and agree on the next step.
Systems Integration and Interface Control	Before we commit, I want to confirm interface control, integration, the owner, and the evidence behind the decision. If interfaces, timing assumptions, version control, and regression testing need governance., I recommend we document the risk and agree on the next step.

High-pressure pushback frames

- I understand the urgency. The risk is that we move faster than the evidence or process supports.
- I am not blocking the goal. I am naming the condition we need before the decision is safe and credible.
- If we accept this risk, we should name the owner, document the assumption, and define the trigger for escalation.
- That may be possible, but not under the current scope, timeline, or approval path.
- Let's separate what we know, what we assume, and what still needs confirmation.

Contrast Pairs

Do not confuse	Useful distinction
requirement vs tradeoff	In requirements and constraints, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
design review vs verification	In design reviews and technical pushback, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
failure mode vs duty cycle	In failure modes and reliability, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
prototype vs sample size	In testing, validation, and data interpretation, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
DFM vs cycle time	In manufacturability and cost engineering, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
safety factor vs liability	In safety factors and compliance, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
field failure vs installation condition	In field issues and customer communication, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
interface control vs configuration management	In systems integration and interface control, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.